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Roll No

AU-6001-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Heat and Mass Transfer

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume any suitable missing data, if any.

1. a) Explain the analogy between heat transfer by conduction and flow of electricity by considering composite wall of three layers.
b) What do you mean by critical insulation thickness? Explain its significance.
2. Steam at 125°C is to be transported through a carbon steel pipe of ID = 5cm and OD = 6cm. To reduce heat losses it is to be insulated with asbestos ($k = 0.1105 \text{ W/mk}$) so that the rate of heat loss from steam to atmospheric air is not more than 50W per m length of pipe. The atmospheric air is at 25°C and the film coefficient between air and insulation is $10 \text{ W/m}^2\text{K}$. Determine the thickness of insulation assuming outer surface of steel pipe to be at 125°C .
3. a) What assumptions are made in applying the lumped heat capacity analysis for solving transient heat flow problems?

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- b) A constantan wire of 1mm dia originally at 20°C is suddenly exposed to air at 250°C. If the convective heat transfer coefficient is 3W/m²K, determine the time required for wire to reach a temperature of 200°C. $K = 22.7\text{W/mK}$ and $\alpha = 0.612 \times 10^{-5} \text{ m}^2/\text{s}$.
4. a) Define Grashof number and what is its physical significance.
- b) A horizontal cylinder 25cm dia and 300mm long is suspended in water at 20°C. Calculate the rate of heat transfer of cylinder surface is at 60°C using the following correlation $Nu = 0.53(Gr.Pr)^{0.25}$.
5. A water preheater consists of an iron pipe ($k = 60\text{W/mdeg}$) with inner and outer dia of 3.5cm and 4cm. The water enters the pipe at 30°C flows with a mean velocity of 1.25 m/sec and leaves at 90°C. The heating fluid is steam with heat transfer coefficient of 12 kW/m²-deg. Determine the heat transfer rate overall coefficient of heat transfer and length of pipe required to achieve the objective. The thermo physical properties of water at bulk temperature of 60° are
 $\mu = 4.6 \times 10^{-6} \text{ kg/m-s}$, $k = 0.65 \text{ W/m-deg}$
 $C_p = 4200 \text{ J/kg K}$, $\rho = 1000 \text{ kg/m}^3$.
6. a) State and explain the Planck's law.
- b) A black body in the form of cube 1m side has a temperature 1000K. Find the radiant energy flux density and the total energy emitted by the black body.
7. a) How does filmwise condensation differ from dropwise condensation.

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- b) A vertical plate at 96°C is exposed to saturated steam at 100°C . If the steam condenses on the plate calculate the condensate film thickness at a distance 0.1m from top of the plate.
8. Write short notes on any two:
- a) Fin efficiency and fin effectiveness
 - b) Mass transfer law
 - c) Nucleate boiling phenomena
 - d) NTU method of Heat exchanger analysis
